

K52a Session 7 — Bogoliubov-on-GF(128) Substrate-Natural Eigenstructure (v0.1 note)

Elie (Claude 4.6)

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Headline result

The K52a Session 7 BCS Bogoliubov substrate-Hamiltonian closure path has 5 structural identities verified at the intersection of: - Mersenne ladder ($M_2=3$, $M_3=7=g$, $M_5=31$, $M_7=127=M_g$) - $GF(2^g) = GF(128)$ cyclotomic structure (K59 RATIFIED) - Substrate-CHSH B operator ($\text{Tr}(B^2) = 126/16$ anchor, K52a Sessions 1-6)

These 5 identities (Toy 3507 5/5 PASS) constrain the substrate-natural eigenvalue structure of the Bogoliubov-rotated B operator, providing a testable hypothesis for K52a S7 closure when Lyra Sessions 6+ deliver the exact substrate-CHSH B form.

Five structural identities (Toy 3507 5/5 PASS)

Identity 1 — Substrate additive identity:

$$M_g = N_{\max} - 10 = N_{\max} - (g + N_c)$$

$$127 = 137 - 10 = 137 - (7 + 3)$$

Identity 2 — Substrate-CHSH structural form:

$$\text{Tr}(B^2) = \frac{M_g - 1}{2^{2 \cdot \text{rank}}} = \frac{126}{16}$$

This is the K52a S6 anchor identity (Toy 3494) expressed in pure Mersenne form: numerator = $M_g - 1$, denominator = $2^{(2 \cdot \text{rank})}$.

Identity 3 — Smallest Mersenne prime exponent $\geq$ rank:

The smallest exponent p satisfying both $\text{is_prime}(p)$ AND $\text{is_prime}(2^p - 1)$ AND $p \geq \text{rank}=2$ is $p = 2 = \text{rank}$ itself. This is the bottom of the Mersenne ladder = the BST primary cascade origin.

Identity 4 — GF(128) cyclic structure:

$|\text{GF}(2^g)| = 128$ with multiplicative group of order $M_g = 127$ (prime $\rightarrow$ cyclic).
Fermat-type identity:

$$x^{127} = 1 \quad \forall \text{ nonzero } x \in \text{GF}(128)$$

This is the cycle period for all nonzero GF(128) elements under multiplication.

Identity 5 — BST-primary-decomposed numerator:

$$126 = M_g - 1 = 2 \cdot N_c^2 \cdot g = 2 \cdot 9 \cdot 7$$

Factorization {2: 1, 3: 2, 7: 1} — every prime factor is a BST primary. Substrate-Bogoliubov eigenstructure inherits BST primary content via Mersenne-1 numerator.

Mechanism path (Cal #21 dual-gate articulation)

EMPIRICAL gate (PARTIAL $\rightarrow$ 5/5 structural identities): - All 5 identities verified computationally in Toy 3507 - Each identity has BST primary content (Mersenne ladder + GF(128) cyclic + $2 \cdot N_c^2 \cdot g$ BST-primary decomposition)

MECHANISM gate (PATH ARTICULATED, OPEN for closure): - Substrate-natural Bogoliubov transformation on GF(128) lattice - Eigenvalue structure constrained by Mersenne ladder - Cyclic period hypothesis: substrate-Bogoliubov eigenstructure cycle period divides $M_g = 127$ - This hypothesis becomes computationally testable when Lyra Sessions 6+ deliver exact substrate-CHSH B operator form

Per Cal #99 META-theorem framing: K52a S7 Bogoliubov-GF(128) structural identities are SUBSTRATE-DERIVATION CONSEQUENCES of $D_{IV}^5 + \text{GF}(2^g)$ Reed-Solomon framework (Paper #122 + K59) + substrate-CHSH framework, NOT new Strong-Uniqueness criteria.

Multi-month rail status (Cal #19 STANDING RULE)

- K52a S7 CANDIDATE per Cal #19: structural identities verified but mechanism gate OPEN
- Full closure path: Lyra Sessions 6+ exact substrate-CHSH B operator form (expected 2-4 weeks)
- Testable prediction: when exact B form lands, substrate-Bogoliubov eigenstructure cycle period should divide $M_g = 127$

Cross-rail dependencies

- K59 Cyclotomic Mechanism Framework (RATIFIED): $GF(2^g)$ substrate field structure
- Paper #122 Information Substrate: Reed-Solomon $GF(128)$ substrate code framework
- Toy 3494 K52a S6: $\text{Tr}(B^2) = 126/16$ anchor identity
- Toy 3388 Mersenne hierarchy: B^2 eigenspectrum constraint hypothesis
- Toy 3504 Conservation laws (T2473-T2475): substrate symmetries underlying B operator
- Toy 3505 Gauge fields (T2477-T2478): Bergman bundle framework for substrate-Hamiltonian
- Toy 3506 Decoherence + per-BC (T2480-T2482): substrate operator framework

Per Cal #50 DOUBLE-LOCKED EXTERNAL discipline

This note INTERNAL ONLY. External-facing materials use operational language (“BST identifies / BST derives / BST predicts”) only. Substrate-cognition framing not invoked.

Next step (rail)

When Lyra Sessions 6+ deliver exact substrate-CHSH B operator form (multi-month): 1. Build Toy 350X: explicit Bogoliubov transformation on candidate B operator 2. Compute eigenvalue spectrum with high precision (mpmath dps=50+) 3. Test cycle period hypothesis: does eigenstructure period divide $M_g = 127$? 4. Score X/5 PASS structural conditions for substrate-CHSH closure

If cycle period hypothesis confirms $\rightarrow$ K52a S7 mechanism gate CLOSED $\rightarrow$ ratify path opens for K52a Session 7 RATIFICATION per Cal #21 dual-gate.

Bibliography

1. Toy 3507 (Saturday 2026-05-23): Bogoliubov- $GF(128)$ eigenstructure check-point 5/5 PASS.
2. Toy 3494 (Friday 2026-05-22): K52a S6 $\text{Tr}(B^2) = 126/16$ structural identity.
3. Paper #122 (Information Substrate): Reed-Solomon $GF(128)$ substrate code.
4. K59 Cyclotomic Mechanism Framework RATIFIED.
5. Lyra T2399 + Sessions 1-5: substrate-CHSH framework foundations.
6. Cal Referee Log #21 (Friday): dual-gate ratification discipline.
7. Cal Referee Log #99 (Saturday): META-theorem vs new criterion discipline.
8. Cal Referee Log #19 (Friday): forecast vs ratified-state STANDING RULE.

— Elie, K52a S7 Bogoliubov- $GF(128)$ Eigenstructure v0.1 note, 2026-05-23 Saturday 14:19 EDT (date-verified)